

A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

- **Type:** Supervised Learning
- **Sub-type:** Regression (predicting a continuous numerical value: house price)

Step-by-Step Logic to Solve It

1. **Understand the Problem & Collect Data**
 - Gather historical house data with:
 - Features: square footage, number of bedrooms, location, etc.
 - Target: house price.
2. **Data Preprocessing**
 - Handle missing values (impute or remove).
 - Convert categorical features (like location) into numerical format using **one-hot encoding** or **label encoding**.
 - Normalize/scale numerical features if needed.
3. **Feature Selection**
 - Identify which features are most relevant to predicting house price.
 - Remove redundant or highly correlated features if necessary.
4. **Split Data**
 - Divide dataset into **training set** and **test set** (e.g., 80%-20%).
5. **Choose & Train Model**
 - Select a regression algorithm:
 - Linear Regression (simple baseline)
 - Decision Tree Regression, Random Forest Regression, or Gradient Boosting for better performance.
 - Train the model on the training data.
6. **Model Evaluation**
 - Predict prices on the test set.
 - Evaluate using regression metrics:
 - **Mean Absolute Error (MAE)**
 - **Mean Squared Error (MSE)**
 - **R² Score**
7. **Make Predictions**
 - Use the trained model to predict house prices for new inputs (e.g., 1500 sq. ft., 3 bedrooms, location “Downtown”).
8. **Deploy**
 - Integrate the model into a real-world system for real-time predictions.

2. A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type: Fraud Detection Model

Problem Type

This is a supervised binary classification problem. The main goal is to train a model to label each transaction as "fraud" or "not fraud" by learning patterns from historical (labeled) data.

Step-by-Step Logic (Personalized for Your Approach)

1. Business Understanding and Problem Framing

- Engage banking SMEs to clarify the definition of fraud relevant to the institution (e.g., card-present vs. card-not-present).
- Establish measurable KPIs: focus on minimizing false negatives (missed fraud) without overwhelming operations with false positives.

2. Data Integration and Preprocessing

- Use your SAP BTP or custom data pipelines to extract, cleanse, and synchronize data from core banking/transactional systems, APIs, and logs.
- Structure the data schema for scalability: consider BigQuery-like architectures if transaction volumes are high.

3. Feature Engineering

- Calculate session-based features, device profiles, geo-spatial anomalies, velocity (rapid spend), and customer behavior deviations using Python (pandas, numpy, etc) and SQL (for big-data scale).
- Utilize AI/ML-specific features—potentially employing OpenCV for any biometric verification or image-based features, as per your interest in computer vision.

4. Handling Imbalanced Data

- Use advanced resampling techniques (SMOTE, ADASYN), class weights, or anomaly detection layering, as fraud classes are usually <5%.
- Automate resampling pipelines using sklearn and imbalanced-learn.

5. Model Selection and Training

- Evaluate interpretable models first (logistic regression, decision trees) for transparency with audit teams, then experiment with ensemble methods (Random Forest, XGBoost) and, if justified, deep learning approaches.
- Split data into train/validation/test using stratified methods to preserve fraud proportion.

6. Model Evaluation and Explainability

- Prioritize recall, precision, F1-score, ROC-AUC; use confusion matrices and cost-sensitive scoring.

- Leverage SHAP/LIME for explainable predictions—this is increasingly expected in regulatory environments.

7. Operational Integration and Deployment

- Implement real-time model scoring in your Python/REST framework or SAP BTP automation. Trigger alerts and case management via APIs.
- Monitor model drift, automate periodic retraining via scheduled jobs/scripts.

8. Continuous Learning & Feedback Loop

- Build an ABAP or Python dashboard for continuous monitoring, feedback from investigations, and automated retraining triggers.
- Collect false positive/false negative feedback for supervised correction.

Example Technologies

- Data: SAP HANA, pandas, SQL, BigQuery (for scale)
- Modeling: Python (scikit-learn, XGBoost, Keras), SAP BTP ML services
- APIs & Automation: Flask/FastAPI, SAP IRPA, custom ABAP endpoints

3) A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

Customer segmentation is an unsupervised machine learning problem where the goal is to group customers based on shared characteristics or behaviors, such as shopping patterns, demographics, or preferences.

Step-by-step approach to solve it:

1. Define Objectives: Clarify the business goal for segmentation, e.g., personalized promotions.
2. Collect Data: Gather customer transaction, demographic, and behavioral data.
3. Prepare Data: Clean data, handle missing values, and normalize features.
4. Feature Engineering: Create meaningful features like purchase frequency, average spend, product preferences, and visit recency.
5. Choose Clustering Algorithm: Apply clustering algorithms such as K-Means, DBSCAN, or hierarchical clustering to form customer groups.
6. Evaluate Segments: Use metrics like silhouette score and interpret cluster characteristics.
7. Deploy: Use segments to design targeted marketing strategies.
8. Monitor and Update: Periodically reassess segments as customer behavior changes.

A company wants to estimate an employee’s salary based on their years of experience, job title, and education level.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type		
This is a supervised regression problem, where the goal is to predict a continuous numeric variable — the employee's salary — based on input features like years of experience, job title, and education level.		
Here is the step-by-step regression problem-solving logic in a generic tabular format:		
Step No	Step Name	Description
1	Define Problem	Understand the goal: predict a continuous numeric value based on input features.
2	Data Collection	Gather a dataset containing features and corresponding numeric target values.
3	Data Preparation	Clean data: handle missing values, outliers; encode categorical variables; normalize features.
4	Feature Engineering	Create or transform features to improve model performance.
5	Split Data	Split dataset into training and testing sets for reliable evaluation.
6	Model Selection	Choose appropriate regression models (e.g., linear regression, decision trees, ensembles).
7	Train Model	Fit the model to training data.
8	Evaluate Model	Use metrics like MSE, MAE, RMSE, R-squared to assess model performance.
9	Tune Model	Optimize hyperparameters through cross-validation or grid search.
10	Deploy Model	Implement the model in production for predicting new data.
11	Monitor & Update	Track model’s performance over time, retrain as needed with new data.

An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

This is a supervised classification problem, specifically binary classification, where the goal is to label each incoming email as either spam or not spam (ham) based on features extracted from the email content and sender details.

Step No	Step Name	Description
1	Understand & Collect Data	Define spam vs. non-spam; collect labeled email data.
2	Data Preprocessing	Clean text (remove punctuation, HTML tags); normalize case; extract metadata features.
3	Feature Extraction	Convert text to numeric features (bag-of-words, TF-IDF, embeddings); include sender details.
4	Split Data	Split labeled data into training and testing sets.
5	Model Selection & Training	Choose and train supervised classifiers (Logistic Regression, Naive Bayes, SVM, etc.).
6	Model Evaluation	Use accuracy, precision, recall, F1-score; focus on minimizing false positives/negatives.
7	Tune Model	Optimize model hyperparameters via cross-validation/GridSearch.
8	Deployment	Integrate the model into email filtering pipelines for real-time spam detection.
9	Monitor & Update	Continuously monitor performance; retrain as spam tactics evolve.

A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

This is a supervised binary classification problem, where the goal is to classify customer reviews into two categories: positive or negative sentiment based on their text content.

Here is the sentiment analysis step-by-step logic formatted as a table:

Step No	Step Name	Description
1	Data Collection	Gather labeled datasets of customer reviews with sentiment tags (positive/negative).
2	Data Preprocessing	Clean text: remove punctuation, stopwords, HTML tags, normalize case, tokenize, lemmatize.
3	Feature Extraction	Convert texts to numerical features using Bag-of-Words, TF-IDF, or word embeddings.
4	Data Splitting	Divide data into training and testing sets for model evaluation.
5	Model Selection & Training	Choose and train classification models like Logistic Regression, Naive Bayes, SVM, or deep learning.
6	Model Evaluation	Assess accuracy, precision, recall, F1-score, and confusion matrix to validate the model.
7	Hyperparameter Tuning	Optimize model parameters using grid search or other techniques.
8	Deployment	Deploy model for automated sentiment classification in production environments.
9	Monitoring & Updating	Monitor model performance and retrain periodically to adapt to new data and sentiment trends.

An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

This is a supervised binary classification problem where the goal is to predict a binary outcome: whether a customer will file a claim (yes/no) in the next year based on historical driving behavior and demographics.

Step-by-Step Solution Approach

Step No	Step Name	Description
1	Problem Definition	Define the objective to predict claim likelihood based on customer data and history.
2	Data Collection	Gather data on driving history, demographics, and prior claims, ensuring quality and relevance.
3	Data Preprocessing	Handle missing data, encode categorical features, normalize numerical features.
4	Feature Engineering	Create new features capturing risk factors such as accident history, vehicle type, and age group.
5	Data Splitting	Split dataset into training and test sets for model validation.
6	Model Selection and Training	Train classification models like logistic regression, random forest, gradient boosting, or SVM.
7	Model Evaluation	Use metrics such as accuracy, precision, recall, F1-score, and AUC-ROC to evaluate performance.

Step No	Step Name	Description
8	Hyperparameter Tuning	Use cross-validation and grid/random search to optimize model parameters.
9	Deployment	Deploy model within insurance platforms for real-time claim risk prediction.
10	Monitoring and Updating	Monitor model performance, retrain with new data to accommodate changes in risk patterns.

A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

This is a clustering or recommendation system problem, where the goal is to group users or movies based on viewing preferences and history to provide personalized movie recommendations.

Step-by-Step Solution Approach

Step No	Step Name	Description
1	Understand & Define the Problem	Clarify the goal: recommend movies based on user viewing patterns and preferences.
2	Data Collection	Gather user viewing history, ratings, and demographic data.
3	Data Preprocessing	Clean data (remove inconsistencies), handle missing data, normalize features.
4	Feature Engineering	Create features like genre preferences, watch frequency, ratings distribution, and viewing time.
5	Data Splitting	Split data into training and validation/test subsets to evaluate recommendation accuracy.
6	Model Selection & Training	Use algorithms like collaborative filtering, content-based filtering, or hybrid methods. Use clustering algorithms (like K-means) if grouping users or movies.
7	Evaluation	Measure performance using metrics such as accuracy, precision, recall, or user satisfaction surveys if applicable.
8	Deployment	Integrate recommendation system into the platform to generate real-time suggestions.

Step No	Step Name	Description
9	Monitoring & Optimization	Monitor user feedback and system performance; update models periodically for better recommendations.

A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

This is a supervised regression problem where the goal is to predict a continuous numerical value (patient recovery time) based on features such as age, medical history, and lifestyle habits.

Step-by-Step Solution Approach

Step No	Step Name	Description
1	Define Problem	Clearly define the objective to estimate patient recovery time post-surgery using patient data.
2	Data Collection	Collect comprehensive patient records including demographics, medical histories, and lifestyle data.
3	Data Preprocessing	Clean data; handle missing values; encode categorical variables; normalize numerical features.
4	Feature Engineering	Extract and create features relevant to recovery such as comorbidities, surgery type, and lifestyle indicators.
5	Data Splitting	Split data into training and testing subsets to assess generalization.
6	Model Selection & Training	Train regression models like linear regression, random forest, gradient boosting, or neural networks.
7	Model Evaluation	Evaluate models using metrics like MSE, MAE, RMSE, and R-squared to measure prediction accuracy.
8	Hyperparameter Tuning	Optimize model parameters via grid search or cross-validation to improve performance.

Step No	Step Name	Description
9	Deployment	Deploy the regression model in clinical decision support systems or hospital management software.
10	Monitoring & Updating	Monitor model accuracy over time and retrain with new patient data to maintain reliability.

A university wants to predict a student’s final exam score based on study hours, attendance, and past academic performance.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type

This is a supervised regression problem where the objective is to predict a continuous numeric variable—students' final exam scores—based on features such as study hours, attendance, and past academic performance.

Step-by-Step Solution Approach

Step No	Step Name	Description
1	Define Problem	Understand and clearly define the objective: predicting exam scores (continuous values) from multiple input features.
2	Data Collection	Collect historical data on students' study hours, attendance records, past academic performance, and final exam scores.
3	Data Preprocessing	Clean data by handling missing values, correcting errors, normalizing or scaling features, and encoding categorical variables.
4	Feature Engineering	Create or transform features if needed to improve model performance (e.g., interaction terms, derived metrics).
5	Data Splitting	Split data into training and testing sets (commonly 70% train, 30% test) to evaluate model generalization.
6	Model Selection & Training	Train regression models such as Linear Regression, Random Forest Regression, Gradient Boosting, or Neural Networks.

Step No	Step Name	Description
7	Model Evaluation	Evaluate model accuracy using metrics like Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).
8	Hyperparameter Tuning	Use techniques like cross-validation and grid or random search to optimize model parameters.
9	Deployment	Deploy the trained model to predict exam scores for new students or scenarios.
10	Monitoring & Updating	Continuously monitor model performance and retrain with new data as needed to maintain accuracy over time.